

PROJECT DESIGN PHASE-I

Proposed Solution

Date: 19 June 2026

Team ID: ----

Project Name: AI-Enhanced Intrusion Detection System

S.No	Parameter	Description
1	Problem Statement (Problem to be solved)	Traditional intrusion detection systems often rely on static rules and signature-based detection methods, which struggle to identify new and evolving cyber threats. Organizations require an intelligent system capable of detecting and classifying network intrusions in real time with high accuracy.
2	Idea / Solution Description	The proposed solution is an AI-Enhanced Intrusion Detection System that uses Machine Learning techniques to analyze network traffic and classify activities as Normal, DoS, Probe, R2L, or U2R attacks. The system utilizes a Random Forest Classifier trained on the NSL-KDD dataset and provides real-time intrusion detection through a Flask-based web application, dashboard, alert logging system, and REST API.
3	Novelty / Uniqueness	Unlike traditional rule-based systems, the proposed solution leverages Artificial Intelligence and Machine Learning for automated threat detection. The system not only detects attacks but also classifies them into multiple attack categories with confidence scores. The integration of a live dashboard, alert management, and API support makes the solution more practical and user-friendly.

S.No	Parameter	Description
4	Social Impact / Customer Satisfaction	The system enhances cybersecurity by helping organizations identify malicious network activities before they cause significant damage. It improves data protection, reduces security risks, and provides users with real-time monitoring and alerts, increasing trust and confidence in network security.
5	Business Model (Revenue Model)	The solution can be offered as a subscription-based cybersecurity service for small and medium enterprises. Revenue can be generated through software licensing, cloud-hosted intrusion monitoring services, premium analytics dashboards, and annual maintenance contracts.
6	Scalability of the Solution	The system is designed with a modular architecture that allows easy integration with cloud platforms, larger datasets, and additional machine learning models. It can scale from small organizational networks to enterprise-level environments by deploying on cloud infrastructure such as AWS, Azure, or Google Cloud Platform.

Conclusion

The AI-Enhanced Intrusion Detection System provides an intelligent, scalable, and cost-effective cybersecurity solution for detecting and classifying network intrusions. By combining machine learning, real-time monitoring, and automated alerting, the project offers a modern approach to protecting networks against evolving cyber threats while supporting future expansion and cloud deployment.